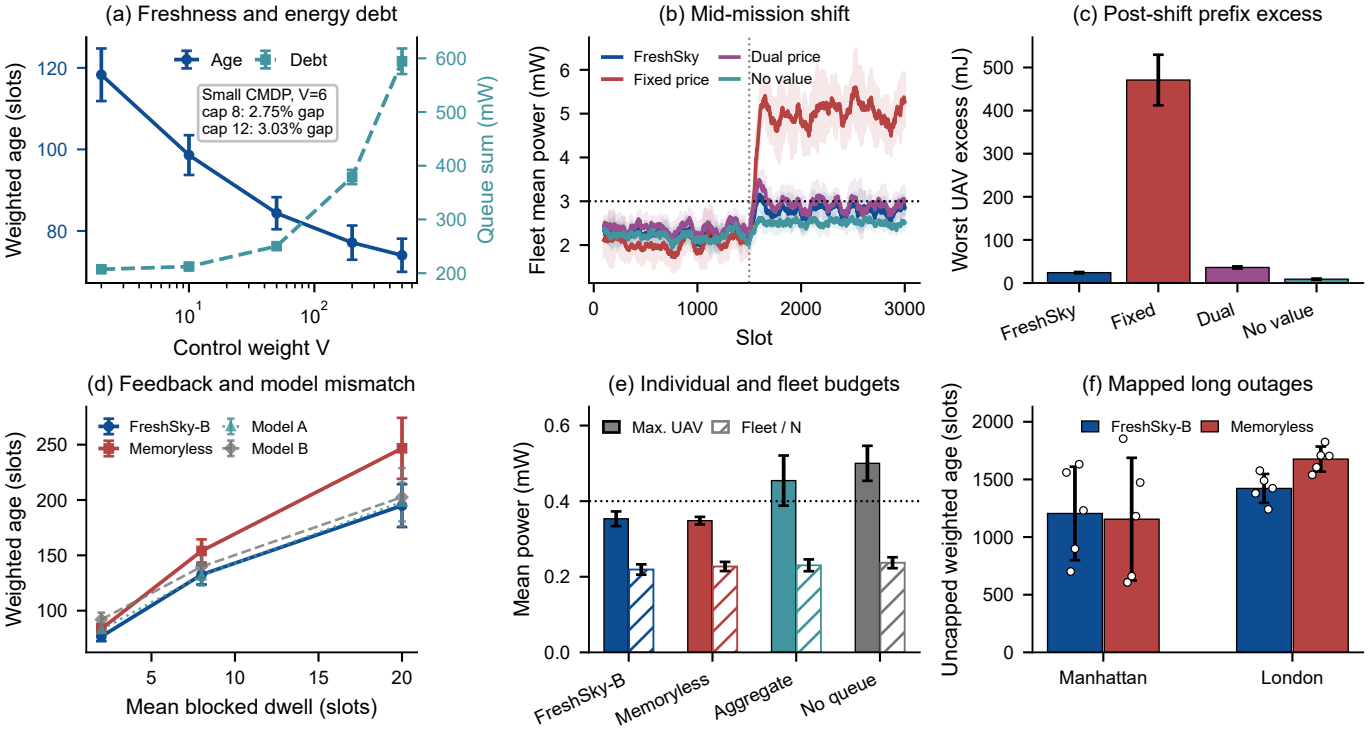

(a) Freshness and energy debt
Age
Debt
Small CMDP, V=6
cap 8: 2.75% gap
cap 12: 3.03% gap
Weighted age (slots)
Queue sum (mW)
Control weight V
(b) Mid-mission shift
FreshSky
Fixed price
Dual price
No value
Fleet mean power (mW)
Slot
(c) Post-shift prefix excess
Worst UAV excess (mJ)
FreshSky
Fixed
Dual
No value
(d) Feedback and model mismatch
FreshSky-B
Memoryless
Model A
Model B
Weighted age (slots)
Mean blocked dwell (slots)
(e) Individual and fleet budgets
Max. UAV
Fleet / N
Mean power (mW)
FreshSky-B
Memoryless
Aggregate
No queue
(f) Mapped long outages
FreshSky-B
Memoryless
Uncapped weighted age (slots)
Manhattan
London